

Ain Shams University

Faculty of Engineering

CHEP – Communications Systems Engineering Program

Spring 2026

ECE359: Digital Image Processing & Multimedia



Formative Midterm

Question One. Consider the following cameraman image. For each of the transformed images (A-D). Draw roughly the linear transformation between input and output pixel values. (4 marks)



Figure 2 Original Cameraman Image

Image A



Image B



Image C

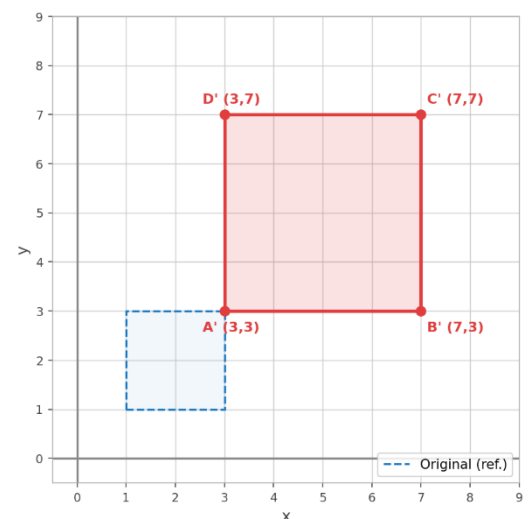


Image D



Figure 1 Images A to D represents transformations

Question Two. Show the 3x3 transformation matrix elements necessary to transform the small square to the larger square in the following figure. (2 marks)



Question Three. Answer the following using the intensity values array which correspond hypothetically to the macro pen tip image. (10 marks)

	c1	c2	c3	c4	c5	c6	c7	c8	
r1	2	3	4	5	5	4	3	2	
r2	3	5	7	8	8	7	5	3	
r3	4	7	10	12	12	10	7	4	
r4	4	7	11	0	12	15	7	4	
r5	5	8	12	14	14	12	8	5	
r6	3	6	9	15	10	0	5	3	Averaging row
r7	2	4	6	7	7	5	3	2	
r8	2	3	4	5	5	4	3	2	

Answers should contain the whole modified row or changed portion of the matrix

- Apply 1D gradient sobel to row r4. What do you notice at the salt and pepper? (1 mark)
- Apply 1D gaussian averaging filter to r6. (1 mark).
- Apply necessary number of suitable filters to remove salt and pepper in rows 4-6. Clearly state which spatial filter you will use, applied location and why? (4 marks)
- Use left image itself as input, right image as target. Perform histogram matching. Clearly show the table, functions, and mapping of the pixels. (4 marks)

Question Four. Miscellaneous questions (4 marks)

- The median filter is used to remove certain amount of noise. Express the noise amount of that type which a 5x5 median filter fails to remove. What other filters can remove this specified noise? (3 marks)
- Clearly explain why discrete histogram equalization fails to reach same results as continuous. (1 mark)

